Teacher Reflection Lab: Using a Big Five–Based Personality Test and Scenario Reflection to Support Novice Teachers' Classroom Communication and Professional Self-Efficacy

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I. Abstract

Novice teachers often enter classrooms with limited opportunities to rehearse the unscripted interpersonal moments of teaching, such as responding to unexpected student questions, adapting a lesson plan, supporting a student who shuts down emotionally, or managing uneven classroom participation. This proposal presents Teacher Reflection Lab, an early-stage simulation and reflection platform designed to support novice and pre-service teachers' classroom communication, uncertainty management, and reflective decision-making. The design is grounded in research on simulation-based teacher preparation, feedback for learning, teacher self-efficacy, and the Big Five personality framework. In the platform, a strengths-based teacher personality profile, informed by Big Five personality dimensions, helps users recognize teaching tendencies, natural strengths, and growth edges. Users are then guided into branching classroom scenarios aligned with six reflective teaching personalities: Community Energizer, Structured Guide, Creative Explorer, Steady Supporter, Reflective Designer, and Quiet Specialist. Each scenario asks users to make consequential teaching choices and then provides an outcome report, a prompt reflection, and an optional "Record the Reflection" feature. Rather than using personality as a fixed label or evaluation tool, Teacher Reflection Lab uses it as an entry point for personalized practice and professional self-awareness. Framed as a design-based proposal, this project has not yet claimed formal learning outcomes. Instead, it aims to investigate the feasibility and practical value of a personality test–simulation–reflection loop to support novice teachers' reflective reasoning, classroom judgment, and confidence during difficult teaching moments. Future evaluation will examine usability, perceived relevance, and evidence of professional noticing in users' written reflections.

II. Proposal Description

Teacher Reflection Lab is a simulation-based reflection platform for novice and pre-service teachers. The platform supports a practice cycle in which users identify a strengths-based teacher personality type, enter a branching classroom scenario, review an outcome report, write a reflection, and decide whether to save that reflection for later review. The project explores whether this personality type–simulation–reflection cycle is feasible and practically useful for helping novice teachers rehearse difficult classroom moments in a low-risk environment.

2.1 Problem and Need

Novice teachers often enter classrooms while still learning how to connect teacher preparation ideals with the realities of school contexts. Stewart and Jansky (2022) describe this transition as a struggle to reconcile theory and practice, especially in schools shaped by standardization, pacing constraints, and complex professional relationships. In their study of novice teacher professional development, participants described challenges related to confidence, expectations, and relationships; one novice teacher explained, "I had a really hard time transitioning into the actual classroom." These findings suggest that novice teachers need more than general encouragement or isolated mentoring. They need structured opportunities to notice, unpack, and respond to the specific struggles they encounter.

Teacher Reflection Lab addresses this need by creating a low-risk simulation and reflection environment for practicing unscripted classroom moments. Our stakeholder interview similarly showed that novice teachers often struggle with communication, unexpected classroom situations, and soft skills, such as responding to questions they do not know how to answer. This design builds on Teacher Moments, which argues that “interactive simulations allow preservice teachers to connect education theory and pedagogy in scaffolded environments” (Thompson et al., 2019). Drawing on Stewart and Jansky’s (2022) concept of “wobble,” the platform treats struggle not as failure, but as a productive starting point for reflection and growth. Through branching scenarios, outcome reports, and prompt reflection, Teacher Reflection Lab supports novice teachers in practicing classroom judgment before facing similar moments in real classrooms.

2.2 Learning Goal

The learning goal of Teacher Reflection Lab is to help novice and pre-service teachers practice reflective classroom judgment in unscripted teaching moments. Users learn to notice classroom tensions, choose communication or instructional moves, interpret how those moves may affect students, and revise their responses for future practice. The broader teaching goal is to help novice teachers build professional self-awareness and self-efficacy by practicing difficult classroom moments in a low-risk environment before encountering similar moments with real students.

2.3 Intended Users and Context

The Teacher Reflection Lab is designed for novice and pre-service teachers in teacher preparation programs or for early-career professional development. It can be used individually or in teacher educator–led workshops as a formative reflection tool.

The platform is not intended for ranking or evaluating teachers. Its purpose is to provide a low-risk practice space where novice teachers can rehearse difficult classroom moments, reflect on their decisions, and build confidence in communication and classroom judgment.

III. Product Design

3.1 Product Description

Teacher Reflection Lab combines a Big Five–based teacher personality test with branching classroom simulations. Users first receive one of six teacher personality types: Community Energizer, Structured Guide, Creative Explorer, Steady Supporter, Reflective Designer, or Quiet Specialist. These types are not used to evaluate teachers, but to help users recognize strengths and growth edges.

Users can then explore scenarios through two pathways. They may choose subject-based or emotion-based scenarios that are aligned with their teacher's personality type. Each scenario focuses on a common novice-teacher challenge, such as uneven participation, adapting a lesson plan, responding to an unexpected question, supporting an upset student, revising instruction based on student work, or making quiet observations visible.

After completing a branching scenario, users receive an outcome report, respond to a reflection prompt, and choose whether to record their reflection. Saved reflections can later be filtered by subject, emotion, or scenario type in their profile.

3.2 User Flow

The current Teacher Reflection Lab prototype follows a continuous learning cycle: personality test → teacher profile →

scenario library → scenario practice → teaching report → reflection. This flow is designed to move novice teachers from self-awareness to low-risk rehearsal to reflective sense-making. The overall design draws on simulation-based teacher preparation, feedback theory, and research on teacher personality and self-efficacy. Instead of treating practice as a one-time assessment, the platform frames each scenario as part of an ongoing process of professional reflection and improvement.

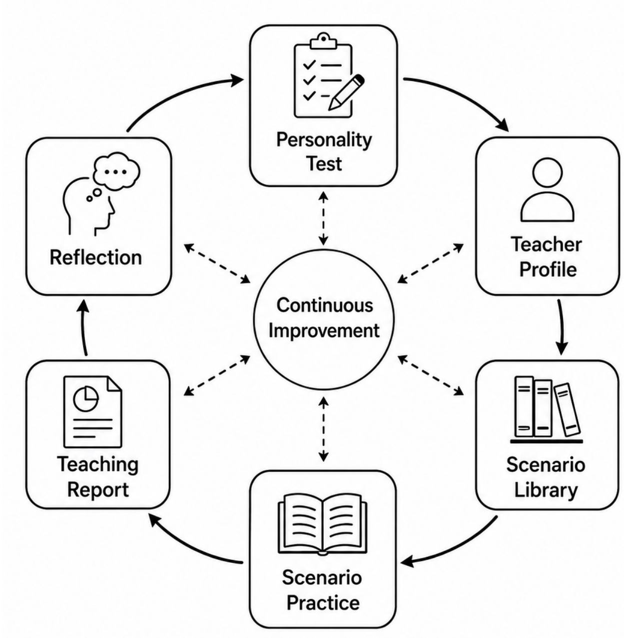

Figure 1. Teacher Reflection Lab User Flow and Continuous Improvement Cycle

3.2.1 Personality Test

The user begins by completing a Big Five–based teacher personality test. The purpose of this step is not to diagnose teacher quality or predict effectiveness, but to help users identify their teaching tendencies in a strengths-based way. The test connects users to one of six teacher personality types: Community Energizer, Structured Guide, Creative Explorer, Steady Supporter, Reflective Designer, or Quiet Specialist.

This step is grounded in Big Five personality theory, which offers a widely used framework for understanding personality traits such as openness, conscientiousness, extraversion, agreeableness, and neuroticism (John & Srivastava, 1999). Research on teacher personality also suggests that personality traits are related to teacher effectiveness, burnout, self-efficacy, work engagement, and job satisfaction (Kim et al., 2019; Perera et al., 2018). In the Teacher Reflection Lab, these ideas are translated into a formative tool for self-awareness rather than evaluation.

3.2.2 Teacher Profile

After completing the test, users receive a teacher profile that summarizes their personality type, strengths, and growth edges. This profile is designed to support professional self-awareness and perceived agency. Perera et al. (2018) found that teacher personality profiles are related to teacher self-efficacy, work engagement, and job satisfaction, suggesting that personality-based reflection can be relevant to how teachers understand their professional capacities. Similarly, Kim et al. (2019) show that teacher personality traits are associated with teacher effectiveness and burnout, which supports the idea that teacher personality can be a meaningful lens for reflection when used carefully.

For this reason, the teacher profile does not present personality as a label or limit. Instead, it frames personality as a starting point for practice. For example, a Community Energizer may be strong at building classroom warmth and participation, but may need practice structuring lively discussions so quieter students can enter. A Structured Guide may be strong at clarity and routines, but may need practice adapting when a lesson does not go as planned. By analyzing both strengths and growth edges, the profile prepares users to enter scenarios with a sense of direction rather than judgment.

This design also supports the platform's broader goal of self-efficacy. Users are not told simply what they lack; they are shown how existing strengths can become resources for growth. The profile, therefore, acts as a bridge between personality awareness and scenario practice.

3.2.3 Scenario Library

After reading teacher profiles, users can explore scenarios through two pathways: subject-based and emotion-based. Subject-based scenarios organize practice around instructional contexts such as discussion, writing feedback, math reasoning, or science inquiry. Emotion-based scenarios are aligned with the user's teacher personality type and focus on affective classroom challenges such as nervousness, uncertainty, frustration, empathy, or classroom energy.

In the early prototype, scenarios were organized mainly as a general classroom library, such as "Math Classroom". However, through playtesting and stakeholder interview feedback, we realized that this structure could create extra cognitive load for teachers with different domain expertise. For example, a novice teacher outside mathematics might focus too much on understanding the content context rather than practicing the intended teaching move. In response, we revised the library so users can choose either subject-based scenarios when the domain feels relevant, or emotion-based scenarios when they want to practice a broader classroom challenge, such as nervousness, uncertainty, frustration, empathy, or classroom energy.

3.2.4 Scenario Practice

In the scenario practice stage, users enter a short branching classroom simulation. Each scenario begins with a realistic classroom moment and asks users to make a sequence of teaching decisions. The choices lead to different branches: some responses close down the learning opportunity, while others create stronger conditions for student participation, emotional safety, or instructional clarity.

This stage is informed by Teacher Moments, which argues that "interactive simulations allow preservice teachers to connect education theory and pedagogy in scaffolded environments" (Thompson et al., 2019). Teacher Reflection Lab applies this idea to classroom-based communication challenges. The goal is not to simulate an entire classroom, but to create focused practice spaces where novice teachers can rehearse difficult moments before encountering them with real students.

3.2.5 Teaching Report

After completing a scenario, users receive a report that explains the outcome of the user's decision path, including what their choices supported, what they missed, and what they might try next time. This design draws on Hattie and Timperley's (2007) feedback framework, which emphasizes that effective feedback helps learners understand their current performance, desired goals, and next steps. In the Teacher Reflection Lab, the report is formative rather than evaluative. It does not simply mark a user's response as right or wrong; it helps users interpret the consequences of teacher moves for classroom climate, student participation, emotional safety, and learning opportunities.

3.2.6 Reflection

After reading the teaching report, users respond to a guided reflection prompt that asks them to revise or transfer their thinking. For example, they may write one sentence they would use in a real classroom or explain how they would respond differently next time. This stage is informed by research and practitioner guidance on reflective practice. Machost and Stains (2023) emphasize that reflection can support educators' professional growth, but it needs clear structures to be usable in practice. Similarly, UW–Madison's Getting Better at Teaching: Start with Reflection argues that becoming better at teaching requires "intentional, reflective practice" and frames reflection as a process of looking back, looking at, and looking ahead: reviewing what worked, noticing what is happening in the moment, and planning future adjustments (University of Wisconsin–Madison School of Education, n.d.).

This framework directly informs the design of the Teacher Reflection Lab. After completing a scenario, users review a teaching report, respond to a structured prompt, and decide whether to record the reflection. By pairing scenario outcomes with guided reflection, the prototype helps novice teachers move from simply seeing a result to interpreting their choices, revising their responses, and preparing for future classroom action.

Users can then choose whether to record the Reflection. This optional step supports user agency. Recorded reflections can later be filtered in the teacher profile by scenario type, making the profile a personal reflection archive rather than a performance tracker.

IV. Research Questions and Methodology

Because Teacher Reflection Lab is currently an early-stage prototype, this proposal does not claim formal learning outcomes. Instead, the next stage of the project will examine whether the personality type–simulation–reflection cycle shows promise for supporting novice teachers' professional growth. Specifically, the planned study will explore changes in users' self-efficacy, the quality of their written reflections, the perceived value of the Big Five–based personality test, and the platform's accessibility and usability.

4.1 Research Questions

This proposed study is guided by four research questions:

RQ1: Self-Efficacy

To what extent does using the Teacher Reflection Lab increase novice teachers' self-reported confidence in responding to unscripted classroom situations?

RQ2: Reflective Growth

To what extent does the scenario–teaching report–reflection loop improve the specificity and actionability of novice teachers' written reflections?

RQ3: Teacher Personality Test

How does the Big Five–based teacher personality test affect users' perceived relevance, self-awareness, and engagement with scenario practice?

RQ4: Accessibility / Usability

What usability challenges emerge as users navigate the personality test, scenario library, teaching report, and reflection recording process?

4.2 Planned Methodology

The proposed study would use a small-scale mixed-methods design with novice or pre-service teachers. Participants would complete the Teacher Reflection Lab cycle: taking the Big Five–based teacher personality test, reviewing their teacher profile, selecting a scenario, completing the branching simulation, reading the teaching report, writing a prompt reflection, and deciding whether to record the reflection.

Data collection would include a pre/post self-efficacy survey, users' written reflection artifacts, post-use interviews, and usability observations. These data would help us examine both the learning promise of the prototype and the practical experience of using it.

Research Question	Data Source	What We Would Examine
RQ1: Self-Efficacy: To what extent does using the Teacher Reflection Lab increase novice teachers' self-reported confidence in responding to unscripted classroom situations?	Pre/post confidence survey	Changes in perceived confidence for handling unscripted classroom situations
RQ2: Reflective Growth: To what extent does the scenario–teaching report–reflection loop improve the specificity and actionability of novice teachers' written reflections?	Written prompt reflections	Specificity, actionability, explanation of classroom tensions, and revised teaching moves
RQ3: Teacher Personality Test: How does the Big Five–based teacher personality test affect users' perceived relevance, self-awareness, and engagement with scenario practice?	Post-use survey and interview	Perceived relevance, self-awareness, personalization, and engagement
RQ4: Accessibility / Usability: What usability challenges emerge as users navigate the personality test, scenario library, teaching report, and reflection recording process?	Usability notes, think-aloud comments, Tobii eye-tracking	Navigation issues, hesitation, confusion, attention patterns, and cognitive load

Table 1: Planned Methodology

Survey data would be analyzed descriptively to identify changes in users' perceived confidence. Written reflections would be coded for evidence of reflective growth, including whether users identify the classroom problem, explain the impact of teacher moves, and propose a revised response. Interview data would be coded thematically around perceived realism, usefulness, emotional safety, and the role of the personality test. Usability data would guide revisions to the interface, scenario library, and reflection-recording process.

V. Expected Contribution and Future Work

5.1 Expected Contribution

Teacher Reflection Lab contributes an early-stage design model for supporting novice teachers' professional growth through a personality type–simulation–reflection cycle. Rather than treating teacher preparation as only the acquisition of strategies, the platform focuses on how novice teachers notice classroom tensions, make decisions under uncertainty, and reflect on the consequences of their responses.

The first contribution is a Big Five–based teacher personality test that functions as a reflective entry point. By connecting users to teacher personality types, the platform helps novice teachers identify potential strengths and growth areas before entering scenario practice.

The second contribution is a set of branching classroom simulations designed around common novice-teacher challenges. These scenarios provide low-risk opportunities for teachers to rehearse difficult classroom moments before encountering similar situations with real students.

The third contribution is the teaching report and reflection system, which moves users from scenario choice to interpretation and revision. Users can choose whether to record their reflection. This supports user agency and keeps the platform formative rather than evaluative.

5.2 Future Work

Future work should begin with a small pilot study with novice and pre-service teachers. This study would collect feedback and research data on users' confidence, reflection quality, perceived usefulness, and usability experience. The results would then be used to improve the next version of Teacher Reflection Lab, including the scenario design, teaching report language, reflection prompts, and interface flow.

Second, the scenario library should be expanded through field studies and interviews with novice teachers, experienced teachers, and teacher educators. The Teacher Reflection Lab should collect real classroom challenges, common emotional tensions, and subject-specific teaching dilemmas from audiences' real experiences.

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